



# HookBazaar

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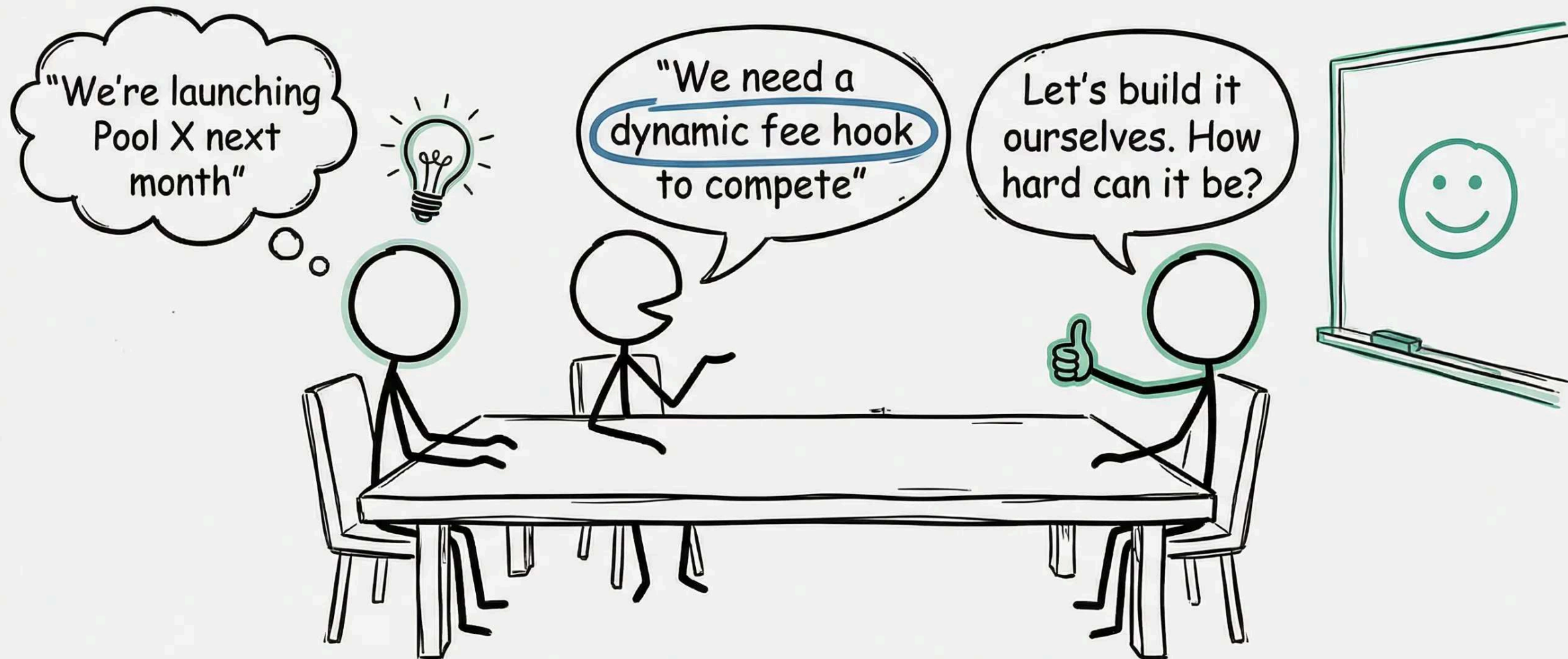
Hook-as-a-Service for Uniswap v4

UHI7 Team: @JMSBPP · @viktor81562 · @iam-Truong

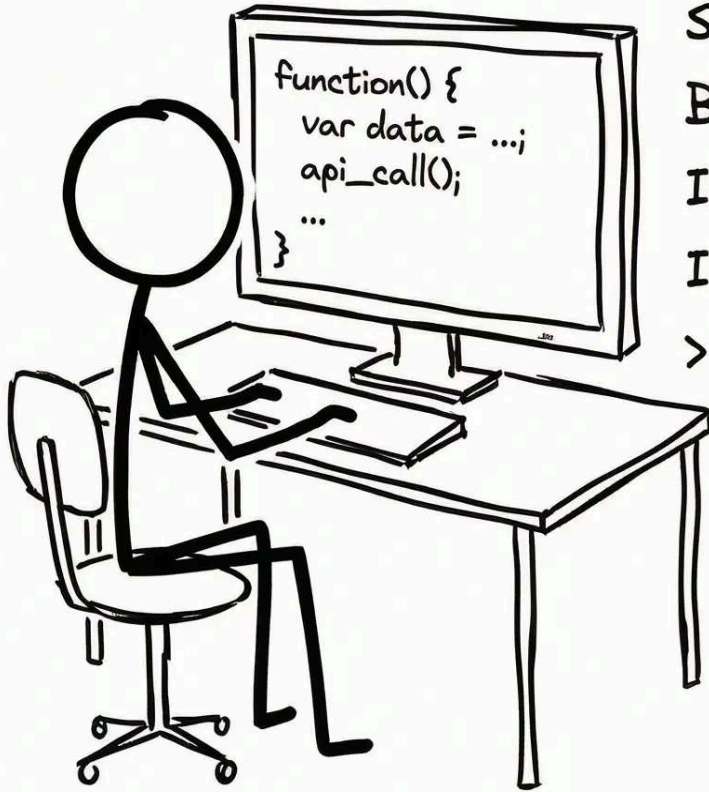
▶ [Watch Demo on Loom](#)

# The Decision

V4



# Building Phase - Development Journey



Setting up infra...  
Building API...  
Integrating AI oracle...  
Implementing callbacks...  
> deploy hook...

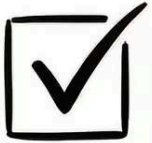
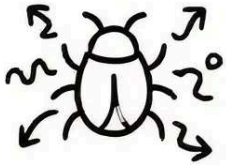
$\$0 \rightarrow \$2,500 \rightarrow \$5,000 \rightarrow \$7,500 \rightarrow \$10,000$

Day 1  $\rightarrow$  Day 7  $\rightarrow$  Day 14  $\rightarrow$  Week 2

$C_{\text{dev}} = \$10,000$  ✓



# Testing Phase - Ensuring Quality



Running fuzzers... ↻

Formal verification... ↻

Integration tests... ✓

Edge cases... ✓

✓ 247 tests passed

COST:

\$10,000 → \$13,000

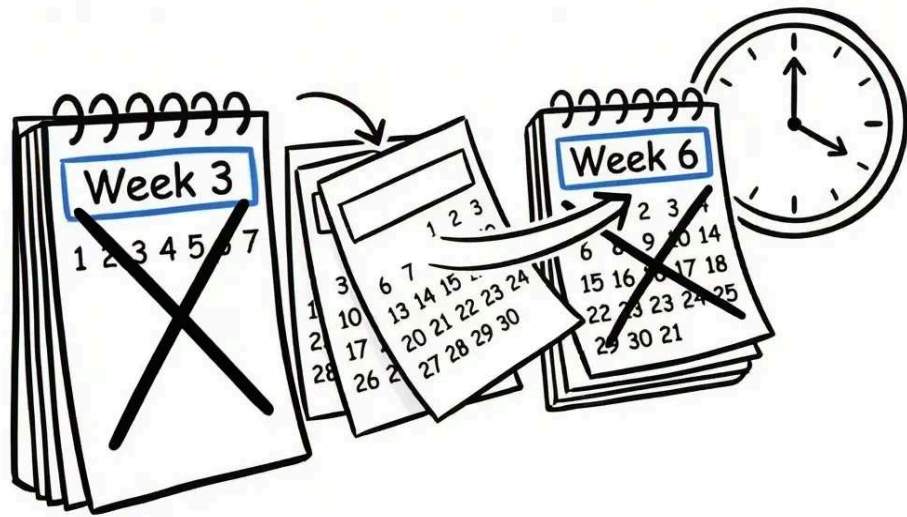
TIME:

Week 2 → Week 3

$C_{\text{test}} = \$3,000$  ✓

Running total: \$13,000

# Audit Phase - Security Review



\$13,000

→ \$15,000

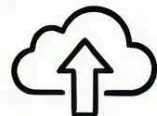
→ \$20,000

→ \$25,000

**\$28,000**

**$C_{\text{audit}} = \$15,000$**

Submitting to  
auditors...



Waiting...



Review in  
progress...



**Running total: \$28,000**

# Total Cost Breakdown

$C_{\text{audit}} = \$15,000$  

$C_{\text{test}} = \$3,000$  

$C_{\text{dev}} = \$10,000$  



$C_{\text{total}} = \$28,000$  !

This was a SIMPLE hook. Complex hooks cost \$50k-\$170k+



## Register Your Protocol

Connect Wallet



Protocol Name: DeFiSuperStars

Submit Registration



Wallet connected



Protocol registered

Welcome to  
Hook Bazaar!

TIME ELAPSED: 30 seconds

[DEMO VIDEO: Registration Flow]

 Search: "dynamic fee"

## Dynamic Fee Hook v2.1 4.8

By: HookDev.eth

Price: \$299

-  Audited by Sherlock
-  47 pools using this hook

[\[View Details\]](#)

[\[Purchase\]](#)

## MEV Shield Hook

 4.9

By: SecureHooks.eth

Price: \$499

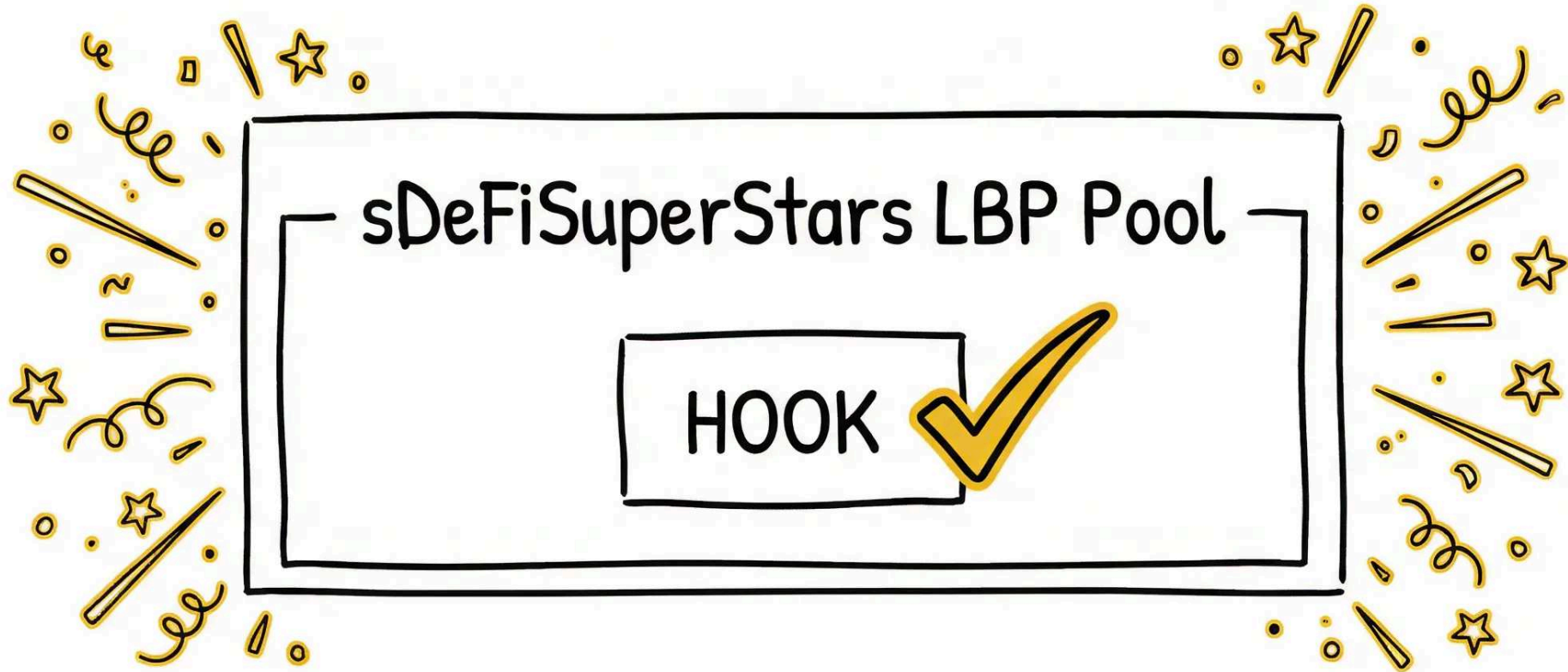
-  Audited by Trail of Bits

TIME ELAPSED: 2 minutes

[\[DEMO VIDEO: Browse Hooks Flow\]](#)



# DEPLOYMENT COMPLETE!



Cost: \$499

Time: 5 minutes

Risk: Minimal  
(pre-audited)

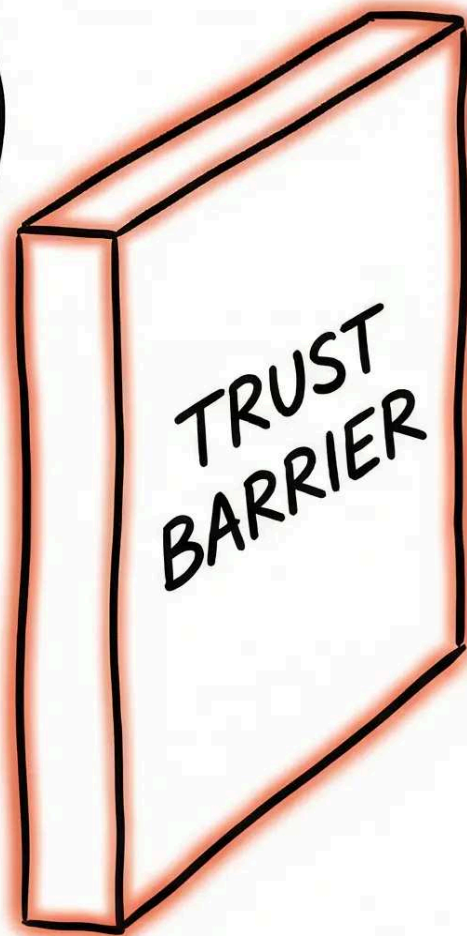
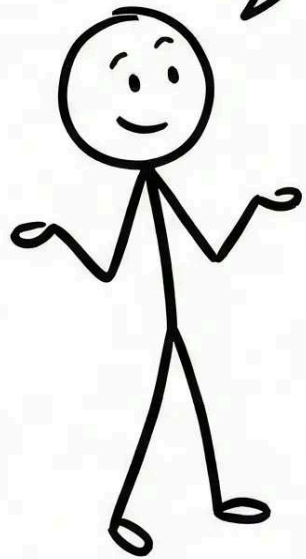
# Meet @eth1000xSuperDev



A Developer's Distribution Challenge



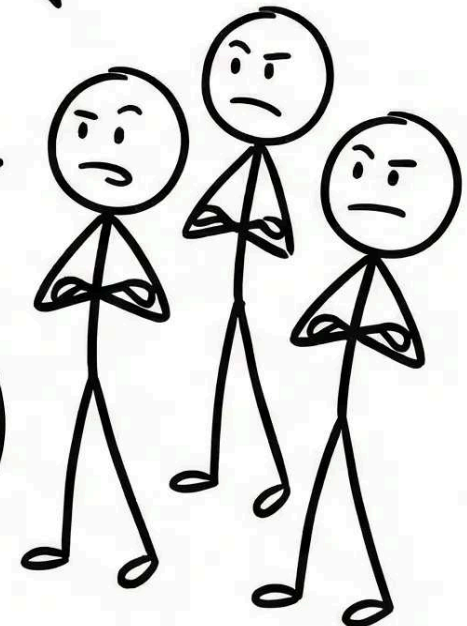
My hook is audited  
and battle-tested!



Who audited it?

Never heard  
of them.

How do we  
know it's secure?



TRUST METER



- Unknown developer: **X**
- Top audit firm: **\$50k+**



~~Free tier  
⇒ no income~~

~~Per-transaction  
⇒ no enforcement~~

~~Flat fee  
⇒ trust needed~~

~~Revenue share  
⇒ no infrastructure~~

How do I even get paid?

## DEVELOPERS



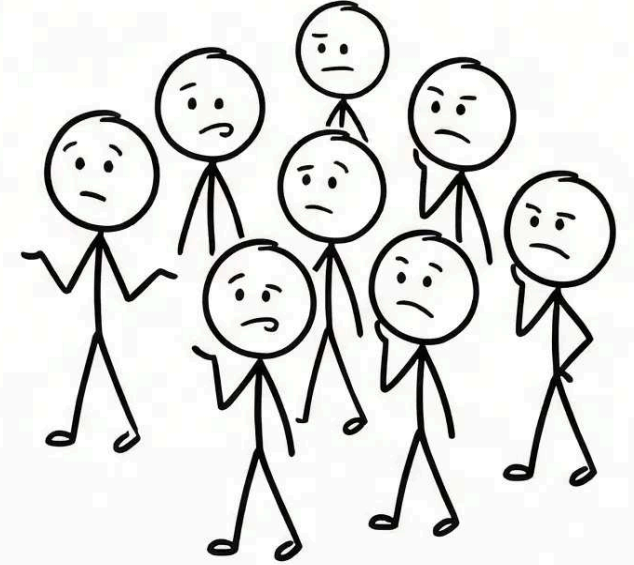
- ✓ Build hooks
- ✗ Can't reach protocols
- ✗ Can't monetize



**NO BRIDGE**

- No discovery
- No trust system
- No payment rails
- No standards

## PROTOCOLS



- ✓ Need hooks
- ✗ Can't find developers
- ✗ Can't verify quality

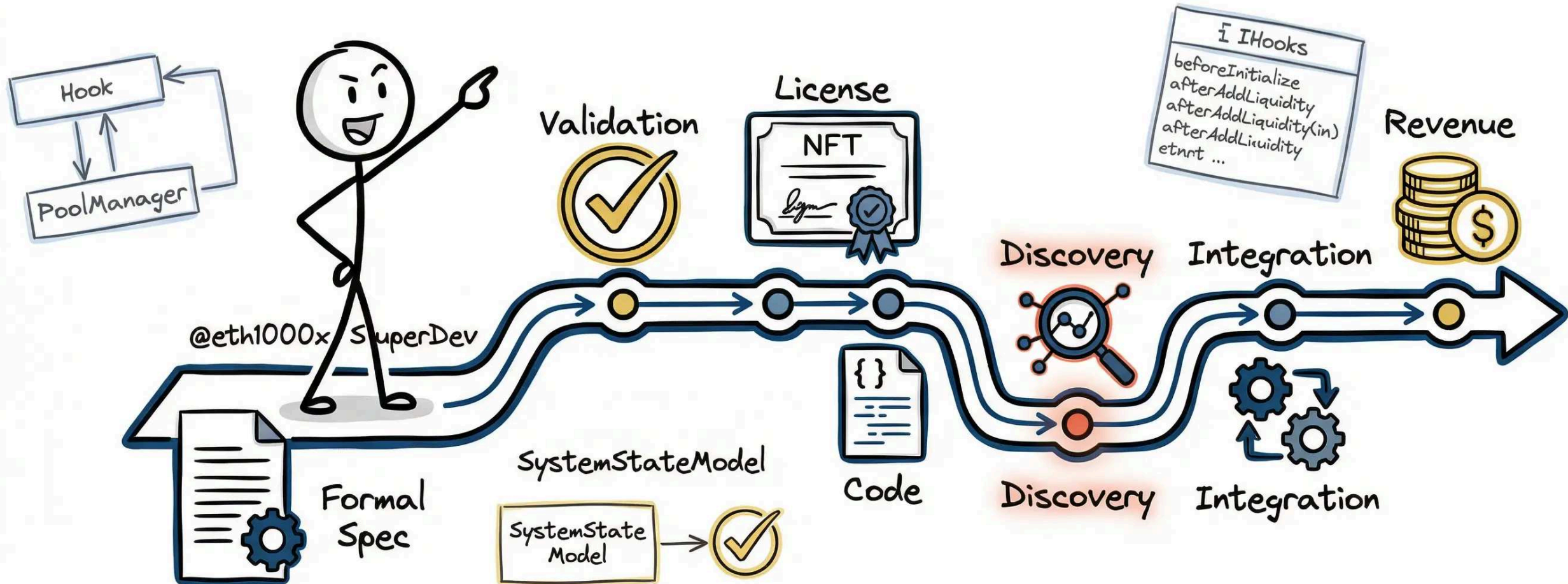
We're both here... but invisible to each other





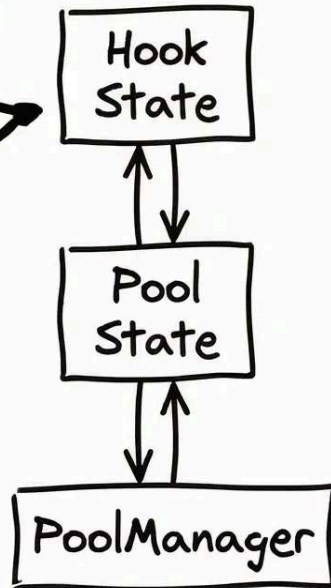
# The Hook Developer Journey: From Specification to Discovery

A Technical Guide to Hook Bazaar's Formal Specification System





Building top-down  
from IHooks to  
achieve my goals



# HOOK SPECIFICATION

Target: IHooks entry points

State Variables:

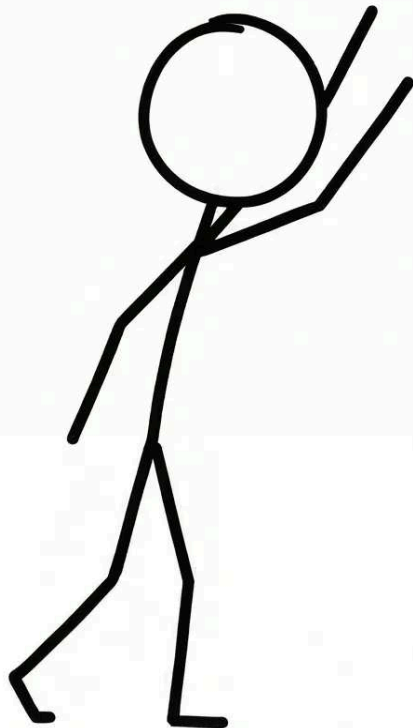
- Maps to `IStateView` canonical vars
- `PoolManager` context interactions

Service Definition:

- `Anti-snipe` protection for LBP pools
- `beforeSwap` callback implementation

Formal Invariants:

- State consistency guarantees
- AMM theory compliance



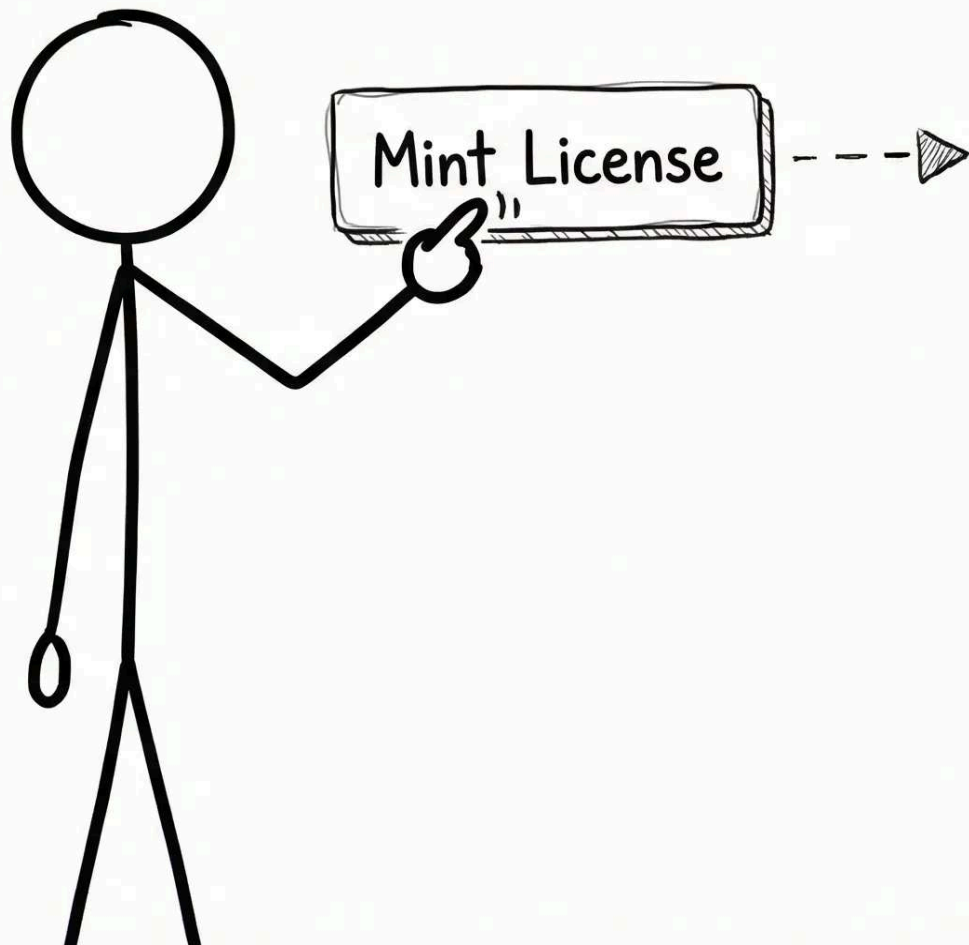
## SystemStateModel Compatibility Check

Validating...

- ✓ State variables mapped correctly
- ✓ IHooks interface compliant
- ✓ PoolManager interactions valid
- ✓ AMM theory constraints satisfied
- ✓ No state conflicts detected

**RESULT: ✓ COMPATIBLE**

Your spec is compatible with  
the SystemStateModel!



## HOOK MINT LICENSE

License ID: #4721

Developer: @eth1000xSuperDev

Spec Hash: 0x7a3f...

Validated:  SystemStateModel

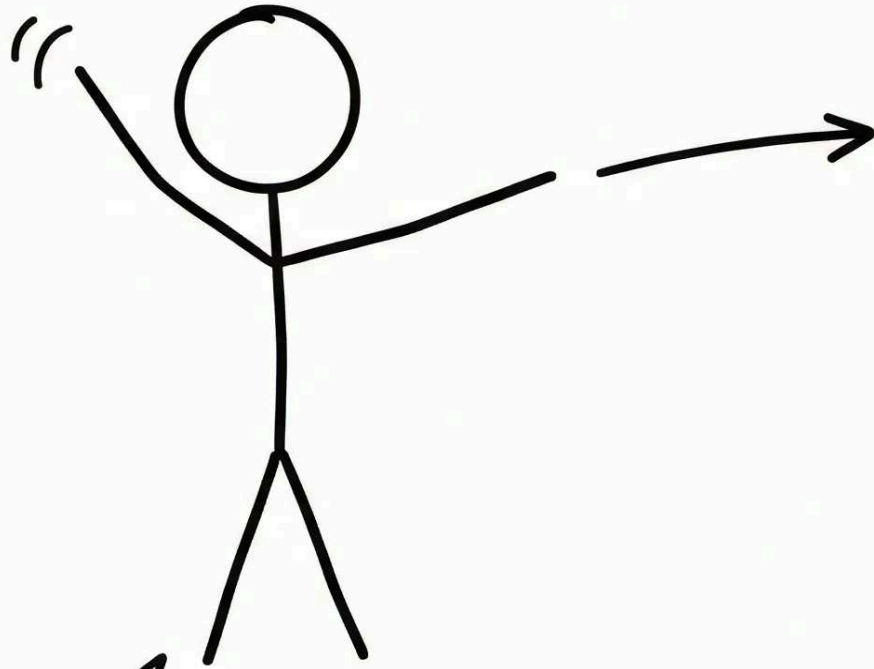
### RIGHTS GRANTED:

-  Submit source code
-  Automatic tagging
-  Metadata generation
-  Marketplace listing

 **Minted!**







The system tagged it automatically and generated the metadata for me!

✓ AntiSnipeHook.sol

### SUBMIT WITH HOOKMINTLICENSE:

#### AUTOMATIC PROCESSING:

- ✓ Code matches spec
- ✓ Compiles successfully
- ✓ Tests pass (247/247)

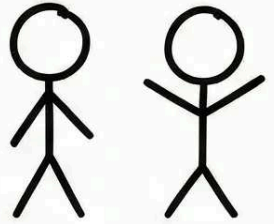
#### AUTO-GENERATED:

- ✓ Tags: [MEV-Protection] [LBP] [beforeSwap]
- ✓ Metadata: Category, compatibility, docs
- ✓ Integration guide created

#### STATUS:

✓ **LIVE ON MARKETPLACE**

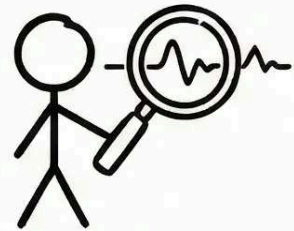
## PROTOCOLS



→ DeFiSuperStars wants to integrate

→ AVS Operator verifies requirements ✓

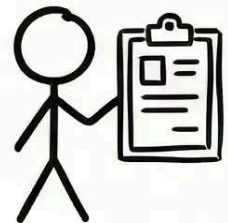
## RESEARCHERS



→ AMM Lab validated HookSpec soundness

→ Formal verification adds credibility ✓

## AUDITORS



→ Sherlock team reviewing for audit

→ Access to code + spec + validation ✓

## DASHBOARD STATS:

Protocol inquiries: 8

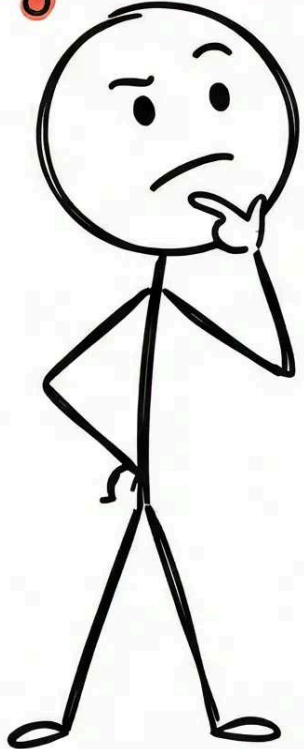
Research citations: 2

Audit requests: 3

Multiple stakeholders, one platform!



# But Wait...



Ok, a marketplace for hooks is great, but...

Once the hook is connected to a pool and deployed on-chain...

Can't someone just DECOMPILE the bytecode?

0x608060405234...

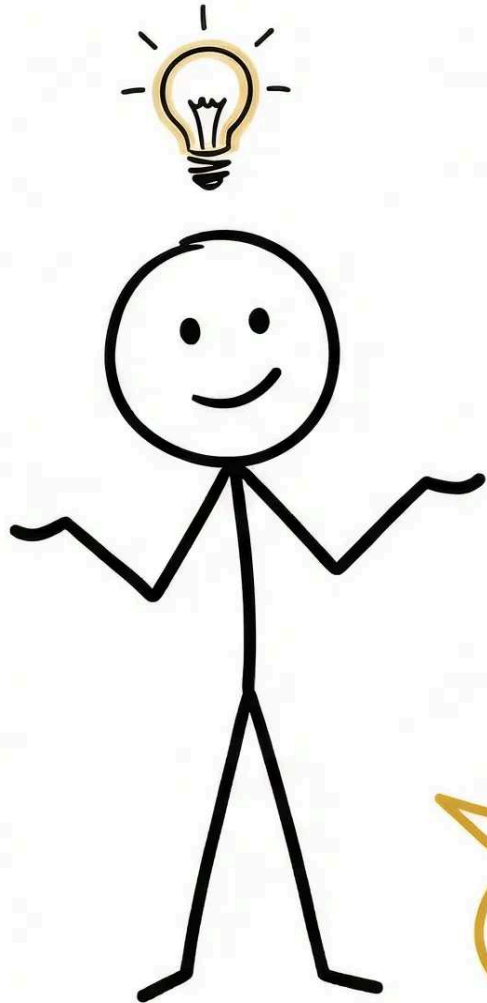
Decompiling tools are getting cheaper.

Why pay subscription when you can just... copy?

**BYTECODE EXPOSED**



# Fhenix FHE Solution



## FULLY HOMOMORPHIC ENCRYPTION

### BEFORE:

- Hook Function Calldata:   
0x608060405234...  
(readable)
- Decompiling Cost:    
\$100-1,000

### AFTER:

- Hook Function Calldata:  [ENCRYPTED]  ?  
(unreadable)
- Decompiling Cost:     
\$1,000,000+

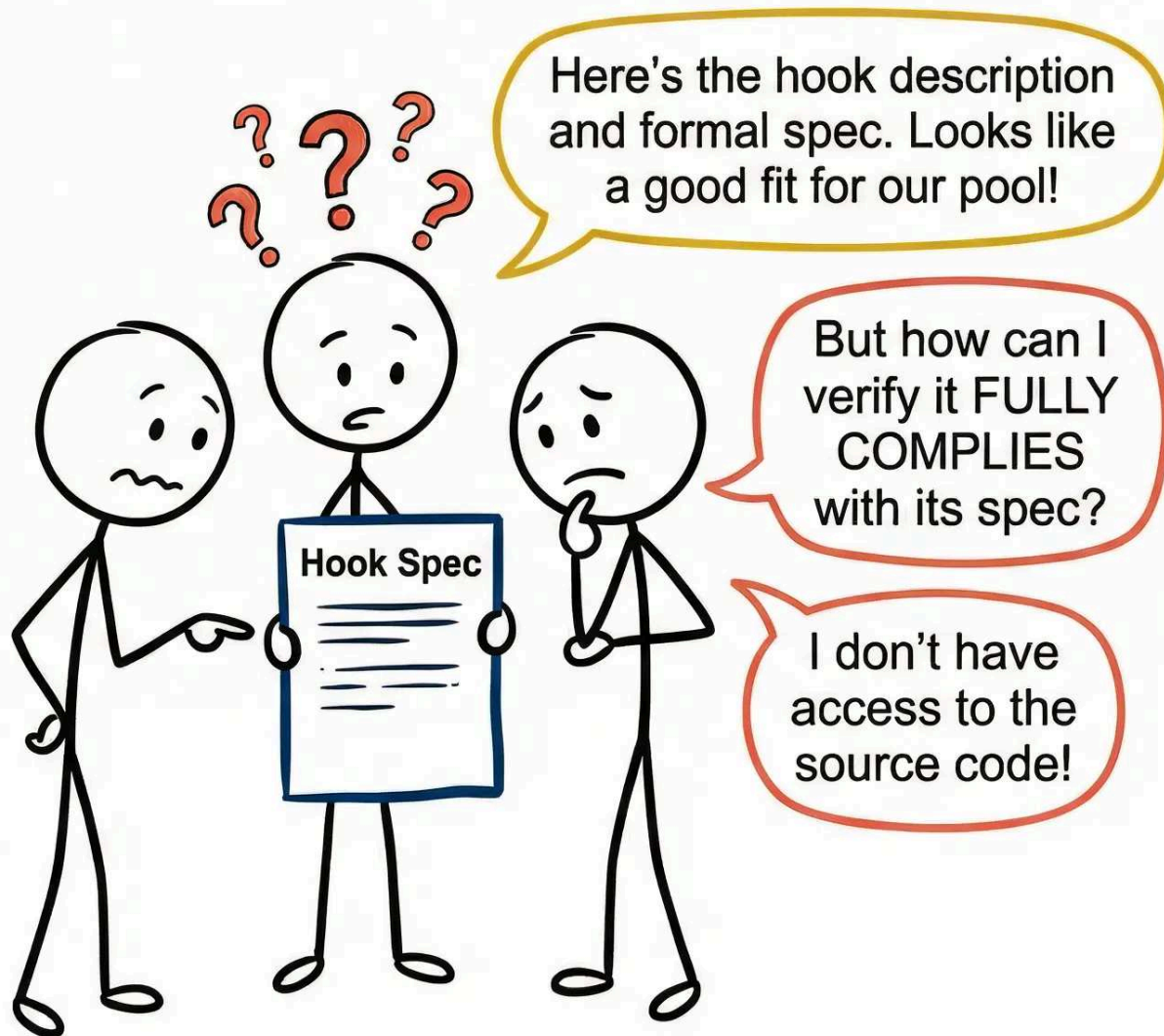
Decompiling cost  
goes to the moon!

Fhenix



[DEMO VIDEO: Fhenix  
FHE encryption flow] 

# Compliance Verification Challenge



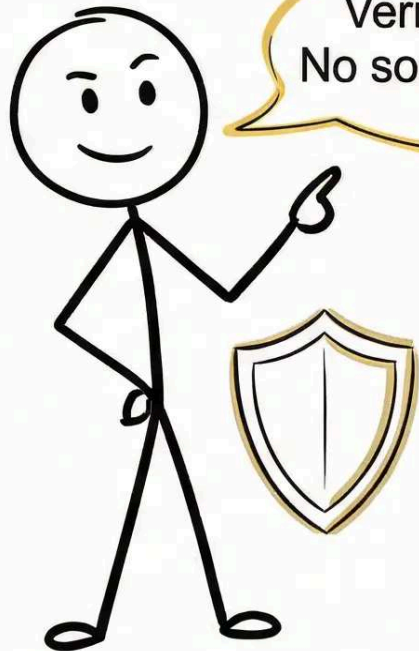
| HOOO SPEC DOCUMENT   | SOURCE CODE   |
|----------------------|---------------|
| ✓ Can read           | ✗ No access   |
| ✓ Looks good         | ? Trust issue |
| ✓ Formal description | ? Compliance? |

**How do I know it does what it says?**



# EigenLayer AVS Attestation

AVS OFF-CHAIN  
OPERATOR



Verified compliance.  
No source code needed.

## SOLUTION

### PROVER ATTESTATION:

- ✓ Compliance Score (detailed metrics)
- ✓ Developer Reputation tracking
- ✓ Audit history integration

### COMPATIBILITY VERIFICATION:

- ✓ Fuzzing integration tests
- ✓ Protocol-specific semantics check
- ✓ Adapter recommendations

RESULT: Trust without source code access!

[DEMO VIDEO:  
AVS Attestation flow]



**EigenLayer**

- Hook submission, AVS operator verification,
- Compliance score generation, Fuzzing test results, Attestation delivery